

SOVEREIGN HEALTH INFRASTRUCTURE

Bloodchain: National Blood Supply Grid & Cryptographic Custody Architecture

An enterprise technical overview of Botswana's decentralised blood supply coordination platform—connecting voluntary donors, laboratory screening, hospital cold storage, and national crisis routing.

Document Reference	Platform URL	Engineering & Origin	Target Sector
BC-ARCH-2026-V2.4 (August 2026)	https://bloodchain.life API: bloodchain-api.onrender.com	Gift Jr Nakedi UniPod, University of Botswana	National Blood Transfusion Service (NBTS) Ministry of Health, Hospital Networks

EXECUTIVE PLATFORM SUMMARY

Bloodchain solves critical blood stock shortages, cold-chain expiration wastage, and delayed maternal postpartum hemorrhage response through a unified 4-pillar software suite. Engineered around the **Strand Ledger Protocol** (tamper-proof cryptographic custody) and the **Torrent Routing Engine** (geo-spatial algorithmic blood balancing), the platform delivers real-time vein-to-vein traceability from voluntary donation to bedside transfusion.

1. National Problem Framing & Strategic Objective

Sub-Saharan national blood services face severe structural supply bottlenecks. In Botswana, the National Blood Transfusion Service (NBTS) targets an annual collection threshold of **35,000+ units**, yet regional facilities face acute deficits. Maternal hemorrhage and acute trauma require immediate blood matching within 30 minutes, yet manual telephone coordination and unlinked hospital reserves lead to critical delays.

Structural Challenge	Current Legacy Reality	Bloodchain Solution Architecture
Maternal Deficits & Acute Trauma	Severe postpartum hemorrhage requires blood within 30 minutes; manual phone coordination causes lethal delays.	Instant multi-facility shortage alerts dispatch targeted notifications to matching universal donors (O- / O+) via Pulse Mobile.
Cold-Chain Expiration & Wastage	Strict 35-42 day red blood cell shelf-life; peripheral clinics lose valuable units due to lack of grid visibility.	Torrent routing algorithm automatically flags nearing-expiry units and coordinates automated rebalancing to high-volume trauma hubs.
Verification & Audit Integrity	Paper logs create chain-of-custody blind spots during inter-hospital transit and serological clearance.	Strand cryptographic ledger seals each step (collection, viral screening, quarantine clearance, cross-match, transfusion) in SHA-256 hashes.

2. Four-Pillar Unified Product Ecosystem

Bloodchain operates as a role-isolated monorepo architecture, providing dedicated interfaces tailored to each stakeholder:

Application	Target Audience	Deployment Mode	Core Capabilities
Pulse (Web & Mobile)	Voluntary Donors & General Public	Android APK (Expo EAS) Web App (React 19 / Vite)	- Digital Sovereign Donor ID with cryptographic QR card - Live national reserve deficit telemetry & shortage alerts - Airtime reward tracking & donation history verification - Seamless lookup / sign-in & consent management
Sanctum	Hospital Blood Bank Staff & Clinicians	Web Terminal (React 19 / Vite)	- Real-time cold storage unit custody tracking - Cross-matching and emergency reservation - Inter-facility transfer initiation and confirmation - Bedside unit release and transfusion logging

Application	Target Audience	Deployment Mode	Core Capabilities
Crucible	Serology & Pathology Lab Technicians	Lab Workstation (React 19 / Vite)	- Mandatory viral screening workflows (HIV, HBV, HCV, Syphilis) - ABO & Rh blood typing validation - Quarantine release authority and test certification - Batch cryptographic signature generation
Vigil	NBTS Command, MoH & National Leadership	Command Center (React 19 / Vite)	- National live inventory grid by hospital & blood type - Real-time critical shortage heat maps and predictive deficits - National transit ledger and inter-district transfer logs - Executive crisis response override tools

3. Technical Architecture & Monorepo Topology

Bloodchain is engineered with a contract-first monorepo pattern using **pnpm workspaces**, ensuring strict type synchronization between the central database, microservices, and client frontends:

Architecture Layer	Technologies	Role & Engineering Implementation
Frontend Applications	React 19, React Native, Expo Router v54, Vite 8, Tailwind CSS	Role-specific responsive web dashboards and compiled native Android APK with offline token caching and biometric/QR card export.
API Microservices	Node.js 24, Express 5, TypeScript 5.9, Zod v4	High-throughput RESTful microservice handling identity, unit lifecycle state transitions, transfer dispatch, and telemetry.
Shared API Spec	OpenAPI 3.0, Orval Codegen, React Query v5	Single source of truth contract (<code>lib/api-spec/openapi.yaml</code>). Generates typed client SDKs and server validation schemas automatically.
Database & Ledger	PostgreSQL 16, Drizzle ORM, Node Crypto (SHA-256)	Relational data store enforcing foreign key constraints, automated migrations, seed orchestrations, and immutable event hashing.
Cloud Infrastructure	Render Cloud Platform, Expo EAS, Custom SSL Domain	Production API (<code>bloodchain-api.onrender.com</code>), edge CDN web deployments on <code>bloodchain.life</code> , and cloud binary builder.

4. Proprietary Core Engines: Strand & Torrent

ENGINE 1: THE STRAND CRYPTOGRAPHIC LEDGER

The **Strand Protocol** guarantees that no blood unit can be transfused without full compliance verification. Every state transition generates an immutable SHA-256 hash incorporating the previous block hash, timestamp, technician identity, temperature log, and lab screening results:

Block Hash = SHA256(PrevHash + UnitID + EventType + LabResults + FacilityID + Timestamp)

Spanning: **Donation Enrolment** → **Collection** → **Viral Clearance** → **Cold Storage** → **Transfer Transit** → **Transfusion**.

ENGINE 2: THE TORRENT SMART ROUTING ALGORITHM

The **Torrent Engine** optimizes national blood allocation by evaluating four dynamic parameters in real time:

- Shelf-Life Expiry Risk:** Prioritizes units nearing the 35-day window for immediate dispatch to high-consumption trauma centers.
- Blood Group Compatibility Matrix:** Protects rare universal reserves (O-negative) by strictly enforcing Rh-iso matching.
- Transit Matrix & Cold-Chain Constraints:** Computes transit time corridors (e.g., PMH Gaborone to Nyangabgwe or Maun General).
- Deficit Severity Score:** Weights maternity wards and trauma surgeries at 3x standard replenishment priority.

5. Data Governance, Consent, & Security Compliance

- Donor Pseudonymization:** National Omang ID numbers and medical histories are decoupled from public QR codes. Public scans display only cryptographic validation tokens and ABO/Rh groups.
- Explicit Applied Telemetry Consent:** The Pulse registration engine incorporates explicit legal consent clauses covering algorithmic testing, shortage notification dispatch, and NBTS clinical reporting.

- **Role-Based Access Control (RBAC):** Laboratory release overrides require certified Crucible technician credentials, preventing unverified or seropositive blood units from entering active hospital inventories.
- **Sovereign Hosting & Isolation:** All health records and ledger events remain under sovereign jurisdiction, with zero dependencies on foreign ad networks or untrusted analytics trackers.

6. Implementation Roadmap & Institutional Deployment

Phase	Milestone & Deployment Scope	Target Infrastructure	Status
Phase 1	Platform Engineering, API Monorepo, & Simulation Grid	Live on https://bloodchain.life (Render + EAS)	COMPLETED (Production Live)
Phase 2	NBTS Gaborone & Princess Marina Hospital (PMH) Pilot	Crucible lab terminal + Sanctum blood bank integration	READY FOR DEPLOYMENT
Phase 3	National Transit Corridor Integration (Nyangabgwe & Maun)	Torrent inter-facility dispatch & cold-chain logger sync	SCHEDULED (Q4 2026)
Phase 4	USSD / SMS Offline Fallback & Africa's Talking Gateway	National network coverage for non-smartphone donors	PLANNED (Q1 2027)

INSTITUTIONAL CONTACT & INQUIRIES

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